

## Calc 2 – Infinite Series

### *Worksheet A*

1. Find the sum  $\sum_{n=0}^{\infty} \left(\frac{2}{3}\right)^n$ .
2. Find the sum  $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$  (telescope).
3. Does  $\sum_{n=1}^{\infty} \frac{1}{n^{3/2}}$  converge? Why?
4. Does  $\sum_{n=1}^{\infty} \frac{n}{2n+1}$  converge? Why?
5. Express  $0.\overline{27} = 0.272727\dots$  as a fraction using a geometric series.

## Calc 2 – Infinite Series

### *Worksheet B*

1. Find the sum  $\sum_{n=1}^{\infty} 5\left(\frac{-1}{4}\right)^{n-1}$ .
2. Find the sum  $\sum_{n=2}^{\infty} \left(\frac{1}{n-1} - \frac{1}{n+1}\right)$  (telescope).
3. Does  $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$  converge? Why?
4. Does  $\sum_{n=1}^{\infty} \cos(n\pi)$  converge? Why?
5. Express  $0.\overline{142857}$  as a fraction.

**Calc 2 – Infinite Series**  
*Answer Key (Worksheets A & B)*

**Worksheet A.**

1. Geometric,  $r = 2/3$ :  $\frac{1}{1-2/3} = \boxed{3}$ .
2. Partial fractions:  $\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$ .  $S_N = 1 - \frac{1}{N+1} \rightarrow \boxed{1}$ .
3.  $p$ -series with  $p = 3/2 > 1$ : **converges**.
4.  $a_n = \frac{n}{2n+1} \rightarrow 1/2 \neq 0$ . By the  $n$ -th term test, **diverges**.
5.  $0.\overline{27} = \sum_{n=1}^{\infty} \frac{27}{100^n} = \frac{27/100}{1-1/100} = \boxed{27/99 = 3/11}$ .

**Worksheet B.**

1. Geometric with  $a = 5$ ,  $r = -1/4$ :  $\frac{5}{1-(-1/4)} = \frac{5}{5/4} = \boxed{4}$ .
2. Partial:  $S_N = (1 - \frac{1}{3}) + (\frac{1}{2} - \frac{1}{4}) + \cdots = 1 + \frac{1}{2} - \frac{1}{N} - \frac{1}{N+1} \rightarrow \boxed{3/2}$ .
3.  $p$ -series with  $p = 1/2 \leq 1$ : **diverges**.
4.  $\cos(n\pi) = (-1)^n$ , does not tend to 0. **Diverges**.
5.  $0.\overline{142857} = \frac{142857}{999999} = \boxed{1/7}$ .